

**Primary HS Code (Silicon Dioxide – Synthetic): 2811.22**

**Natural Silica (If applicable): 2506**

### **SILICA (COSMETIC GRADE)**

Cosmetic-grade silica is highly purified silicon dioxide (SiO<sub>2</sub>) used in personal care, skincare, decorative cosmetics, oral care, and haircare formulations. It functions primarily as an absorbent, anti-caking agent, thickener, viscosity controller, mattifying agent, and texture enhancer.

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### **PRECIPITATED SILICA (COSMETIC GRADE)**

Precipitated silica is synthetically produced through wet chemical processing. It has controlled porosity and high absorption capacity.

It is widely used for oil control and texture enhancement in skincare and powders.

#### **Precipitated Silica – Cosmetic Specification**

| Parameter               | Typical Range               |
|-------------------------|-----------------------------|
| SiO <sub>2</sub> Purity | ≥ 98%                       |
| Surface Area (BET)      | 100 – 250 m <sup>2</sup> /g |
| Loss on Drying          | ≤ 6%                        |
| pH (5% Suspension)      | 6 – 8                       |
| Heavy Metals (Pb)       | ≤ 10 ppm                    |
| Microbial Count         | Cosmetic compliant          |
| Particle Size           | 5 – 50 microns              |

#### **Applications:**

- Face powders
- Foundations
- Oil-control products
- Toothpaste

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### **FUMED SILICA (COSMETIC GRADE)**

Fumed silica is produced via flame hydrolysis, resulting in ultra-fine nano-sized particles with very high surface area. It acts as a thickener and rheology modifier.

### Fumed Silica – Cosmetic Specification

| Parameter               | Typical Range               |
|-------------------------|-----------------------------|
| SiO <sub>2</sub> Purity | ≥ 99.8%                     |
| Surface Area            | 150 – 400 m <sup>2</sup> /g |
| Particle Size           | 7 – 40 nm                   |
| Loss on Ignition        | ≤ 2%                        |
| Heavy Metals            | Within cosmetic limits      |

#### Applications:

- Cream thickening
  - Gel stabilization
  - Long-wear cosmetics
  - Silicone-based products
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### MICROSPHERE SILICA (SPHERICAL SILICA)

Spherical silica particles provide soft-focus and smoothing effects in cosmetic formulations. They improve texture and reduce shine.

### Spherical Silica – General Specification

| Parameter        | Typical Range       |
|------------------|---------------------|
| SiO <sub>2</sub> | ≥ 99%               |
| Particle Size    | 3 – 20 microns      |
| Oil Absorption   | Moderate to high    |
| Appearance       | Free-flowing powder |

#### Applications:

- Blurring primers
- Matte foundations
- Setting powders
- Skincare emulsions

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## COLLOIDAL SILICA (COSMETIC GRADE)

Colloidal silica is a liquid dispersion of nano-silica particles suspended in water. It is used for stabilization and polishing applications.

### Colloidal Silica – Cosmetic Specification

| Parameter                | Typical Range            |
|--------------------------|--------------------------|
| SiO <sub>2</sub> Content | 20% – 50%                |
| Particle Size            | 5 – 100 nm               |
| pH                       | 3 – 10 (grade dependent) |
| Appearance               | Clear to milky liquid    |

#### Applications:

- Film stabilizer
- Cosmetic emulsions
- Specialty formulations